

# KILLARA HIGH SCHOOL



## ASSESSMENT TASK 1 2009 Mathematics Extension 1

**STUDENT NUMBER:** \_\_\_\_\_

**TIME ALLOWED:** 70 minutes

**INSTRUCTIONS:** This paper consists of 4 questions  
ANSWER all four questions  
Start each question in a new booklet  
Marks may not be given for untidy and poorly arranged work.  
All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

- applies appropriate techniques from the study of calculus and series to solve problems (H5)
- uses the derivative to determine the features of the graph of a function(H6)
- uses the relationship between functions and their derivatives(HE4)
- evaluates mathematical solutions to problems and communicates them in an appropriate form(HE7)
- solves problems involving permutations and combinations, and polynomials (PE3)

| OUTCOMES                                       | MARKS |
|------------------------------------------------|-------|
| 1. Polynomials                                 | /14   |
| 2. Permutations and Combinations               | /10   |
| 3. Geometrical Applications of Differentiation | /10   |
| 4. Mathematical Induction                      | /6    |
|                                                | /40   |

*This assessment task constitutes 10% of the Higher School Certificate Course Assessment.*

**QUESTION 1 (14 Marks)    Start a new booklet**

- (a) For the polynomial  $H(x) = 2x^3 - x^2 - 8x + 4$ ,
- (i) find the remainder when  $H(x)$  is divided by  $x + 3$ . 1
  - (ii) Show that  $x - 2$  is a factor of  $H(x)$ . 1
  - (iii) Find the intercepts that the graph of  $y = H(x)$  makes with the  $x$  axis. 2
- (b) The polynomial  $x^3 - 2x^2 + kx + 24 = 0$  has roots  $\alpha, \beta$  and  $\gamma$ .
- (i) Find the value of  $\alpha + \beta + \gamma$  1
  - (ii) Find the value of  $\alpha\beta\gamma$  1
  - (iii) It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of  $k$ . 2
- (c) The polynomial  $P(x) = 2x^4 - ax^2 + bx - 1$  is exactly divisible by  $x + 1$  and has a remainder of  $-3$  on division by  $(x - 2)$ . Find the values of  $a$  and  $b$ . 3
- (d) Find the quotient,  $Q(x)$ , and the remainder,  $R(x)$ , when the polynomial  $P(x) = x^4 - x^2 + 1$  is divided by  $x^2 + 1$ . 3

**QUESTION 2 (10 Marks)    Start a new booklet**

- (a) The mathematics staff currently consists of 6 male and 7 female teachers.
- (i) A team of 4 males and 5 females is to be chosen. 1  
Write an expression for the number of ways this can be done.
- (ii) How many teams of 9 be chosen which contain at 2  
least 5 female teachers?
- (b) How many arrangements of the letters ~~SCOOBIEDOO~~ are possible? 2
- (c) Six students are to be seated in a row on the stage for an assembly.  
How many ways can they be placed if:
- (i) there are no restrictions on where they sit? 1
- (ii) two particular students insist on sitting next to each other? 1
- (iii) two particular students refuse to sit next to each other? 1
- (d) Solve  ${}^nC_2 = 66$  2

**QUESTION 3 (10 Marks) Start a new booklet**

- (a) The doctors report said 'Although the patient's temperature is still rising, attempts to bring it down to normal are having their effect.

The patient's temperature is  $C^\circ$ .

- (i) Jo said ' $\frac{dC}{dt} > 0$ '. Is Jo correct? Explain your answer. 1

- (ii) What conclusion can Jo draw about  $\frac{d^2C}{dt^2}$ ? 1  
You must explain your answer.

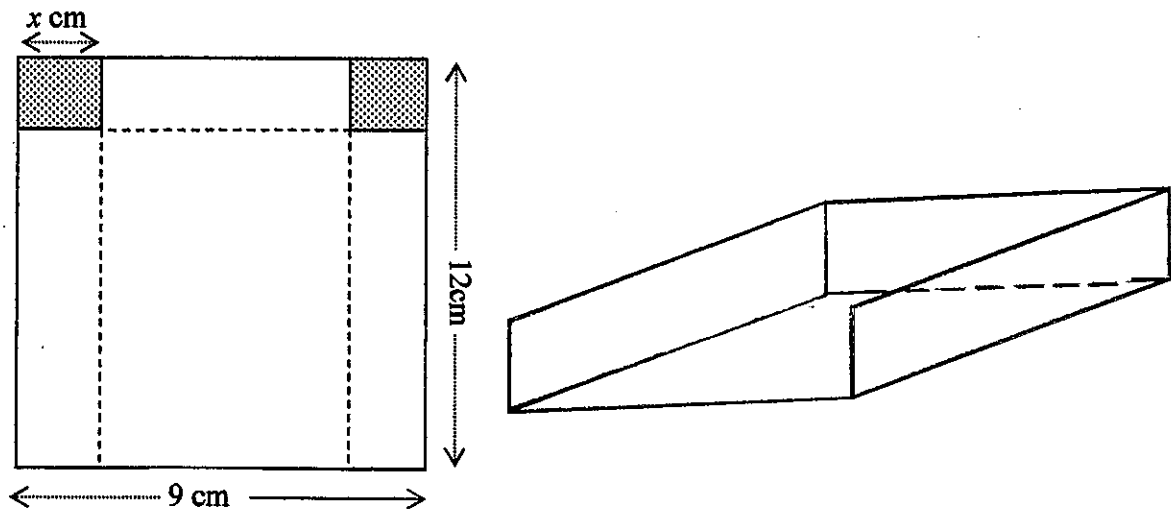
- (b) A rectangle of cardboard measures 12cm by 9cm. From two corners, squares of side  $x$  cm are removed, as shown in the diagram below.

The remainder is folded along the dotted lines to form a tray.

- (i) Show that the volume  $V \text{ cm}^3$ , of the tray is given by 1

$$V = 2x^3 - 33x^2 + 108x$$

- (ii) Find the maximum possible volume of the tray. 3



- (b) Consider the function  $f(x) = \frac{x}{x+2}$
- (i) Show that  $f'(x) > 0$  for all  $x$  in the domain 2
- (ii) State the equation of the horizontal asymptote of  $y = f(x)$  1
- (iii) Without using any further calculus, sketch the graph of  $y = f(x)$  1

**QUESTION 4 (6 Marks)    Start a new booklet**

- (a) Prove by mathematical induction that for a positive integer  $n$ :

**4**

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

- (b) If the following statement is true for  $S_k$ , show that it is also true for  $S_{k+1}$ .

**2**

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k, \text{ for all } k \geq 1.$$

**END OF EXAMINATION**

2009

## Marking Scheme

1 a) (i)  $H(x) = 2x^3 - x^2 - 8x + 4$

$$H(-3) = 2(-3)^3 - (-3)^2 - 8(-3) + 4$$

$$H(-3) = -35$$

$\therefore$  remainder is -35

Award 1 for correct answer

(ii) If  $x-2$  is a factor,  $H(2) = 0$

$$H(2) = 2(2)^3 - 2^2 - 8(2) + 4$$

$$= 16 - 4 - 16 + 4$$

$$= 0$$

$H(2) = 0$ ,  $\therefore x-2$  is factor

MUST HAVE

$x-2$  is factor,  $H(2) = 0$

& full substitution

[because it's a show question]

(iii)  $H(x) = (x-2)(2x^2 + 3x - 2)$

$$= (x-2)(2x-1)(x+2)$$

Intercepts occur when  $H(x) = 0$

$\therefore$  intercepts are  $x=2$ ,  $x=\frac{1}{2}$ ,  $x=-2$

Award 1 for correct answer

b)  $x^3 - 2x^2 + kx + 24$

(i)  $\alpha + \beta + \gamma = -\frac{b}{a}$   
 $= 2$

Award 1

(ii)  $\alpha\beta\gamma = -\frac{d}{a}$   
 $= -24$

Award 1

(iii) Let roots be  $\alpha, -\alpha, \beta$

$$\therefore \alpha + (-\alpha) + \beta = 2$$

$$\beta = 2$$

Award 1 for correct value of 3<sup>rd</sup> root

As 2 is a root,  $2^3 - 2 \times 2^2 + k \times 2 + 24 = 0$

$$8 - 8 + 2k + 24 = 0$$

$$k = -12$$

Award 1 for correct value of  $k$   
 cfpa.

$$c) P(x) = 2x^4 - ax^2 + bx - 1$$

As  $x+1$  is a factor,  $P(-1) = 0$

$$2(-1)^4 - a(-1)^2 + b(-1) - 1 = 0$$

$$2 - a - b - 1 = 0$$

$$a + b = 1 \quad \text{--- (1)}$$

Award 1

As  $-3$  is remainder,  $P(2) = -3$

$$2(2)^4 - a(2)^2 + b(2) - 1 = -3$$

$$32 - 4a + 2b - 1 = -3$$

$$-4a + 2b = -34$$

$$-2a + b = -17 \quad \text{--- (2)}$$

Award 1

$$\textcircled{1} - \textcircled{2} \quad 3a = 18$$

$$a = 6 \quad \text{--- (*)}$$

$$\textcircled{*} \rightarrow \textcircled{1} \quad 6 + b = 1$$

$$b = -5$$

$$\therefore a = 6, b = -5$$

Award 1 for correct sol'n.

$$\begin{array}{r} \phantom{(d)} \quad \quad \quad x^2 - 2 \\ x^2 + 1 \overline{) x^4 - x^2 + 1} \\ \underline{x^4 + x^2} \phantom{+ 1} \\ -2x^2 + 1 \\ \underline{-2x^2 - 2} \\ 3 \end{array}$$

$$\therefore x^4 - x^2 + 1 = (x^2 + 1)(x^2 - 2) + 3$$

$$\therefore \text{quotient } Q(x) = x^2 - 2$$

Award 1

$$\text{remainder } R(x) = 3$$

## Question 2

a) (i)  ${}^6C_4 \times {}^3C_5$

Award 1 for  
correct  
EXPRESSION

(ii) At least 5 females means  
5 females or 6 females or 7 females

ie  ${}^7C_5 \times {}^5C_4 + {}^7C_6 \times {}^6C_3 + {}^7C_7 \times {}^7C_2$

Award 1 for  
CNE

$= 315 + 140 + 15$

$= 470$

Award 1 for  
correct answer

b) SCOOBLED00 : Arrangements  $= \frac{10!}{4!}$

Award 2 for  $\frac{10!}{4!}$

$= 151200$

Award 1 for  
 $\frac{10!}{4!}$  or division by  
 $4!$

c) (i)  $6! = 720$

Award 1

(ii)  $5! \times 2 = 240$

Award 1

(iii)  $6! - 2 \times 5! = 480$

Award 1

d)  ${}^nC_2 = 66$        $\frac{n!}{2!(n-2)!} = 66$        $\therefore n > 2$

Award 2 for  
 $n=12, n > 2$   
[must have  $n > 2$ ]

$\frac{n(n-1)}{2} = 66$

$n^2 - n - 132 = 0$

$(n-12)(n+11) = 0$

$n = 12 \text{ or } -11$

$n > 2, \therefore n = 12$

Award 1 for  
 $\frac{n!}{2!(n-2)!} = 66, n > 2$

### Question 3.

a) (i) To is correct.  $\frac{dC}{dt} > 0$  means that

Award 1

as time increases, so does temperature

(ii)  $\frac{d^2C}{dt^2} < 0$  because as temperature is increasing, it is increasing at a decreasing rate

Award 1

$$\begin{aligned} \text{b) (i) } V &= (12 - x)(9 - 2x)x \\ &= (108 - 33x + 2x^2)x \\ V &= 2x^3 - 33x^2 + 108x \end{aligned}$$

Award 1

MUST HAVE ALL STEPS

(ii) MAX VOLUME when  $\frac{dV}{dx} = 0$

$$\begin{aligned} \frac{dV}{dx} &= 6x^2 - 66x + 108 \\ &= 6(x^2 - 11x + 18) \\ &= 6(x - 9)(x - 2) = 0 \end{aligned}$$

$$\frac{dV}{dx} = 0 \Rightarrow x = 9 \text{ or } x = 2$$

Award 1 for  $x = 9$  or  $x = 2$

$$\frac{d^2V}{dx^2} = 12x - 66$$

$$\text{When } x = 9, \frac{d^2V}{dx^2} = 12 \times 9 - 66 \leftarrow$$

$$\frac{d^2V}{dx^2} > 0 \therefore \text{minimum}$$

$$\text{When } x = 2, \frac{d^2V}{dx^2} = 12 \times 2 - 66 \leftarrow$$

$$< 0$$

Award 1 for testing SP's  
MUST SHOW SUBSTITUTIONS

$\therefore$  Maximum when  $x = 2$

$$\begin{aligned} \text{Maximum volume is } (12 - 2)(9 - 4)2 \\ &= 10 \times 5 \times 2 \\ &= 100 \text{ cm}^3 \end{aligned}$$

Award 1 for correct answer

e) (i)  $f(x) = \frac{x}{x+2}$

(i)  $f'(x) = \frac{(x+2) \cdot 1 - x \cdot 1}{(x+2)^2}$   
 $= \frac{2}{(x+2)^2}$

As  $(x+2)^2$  is always  $> 0$

$\frac{2}{(x+2)^2} > 0$

$\therefore \frac{dy}{dx} > 0 \therefore$  increasing

Award 1 for derivative.

Award 1 for explanation AND

$\frac{dy}{dx} > 0$

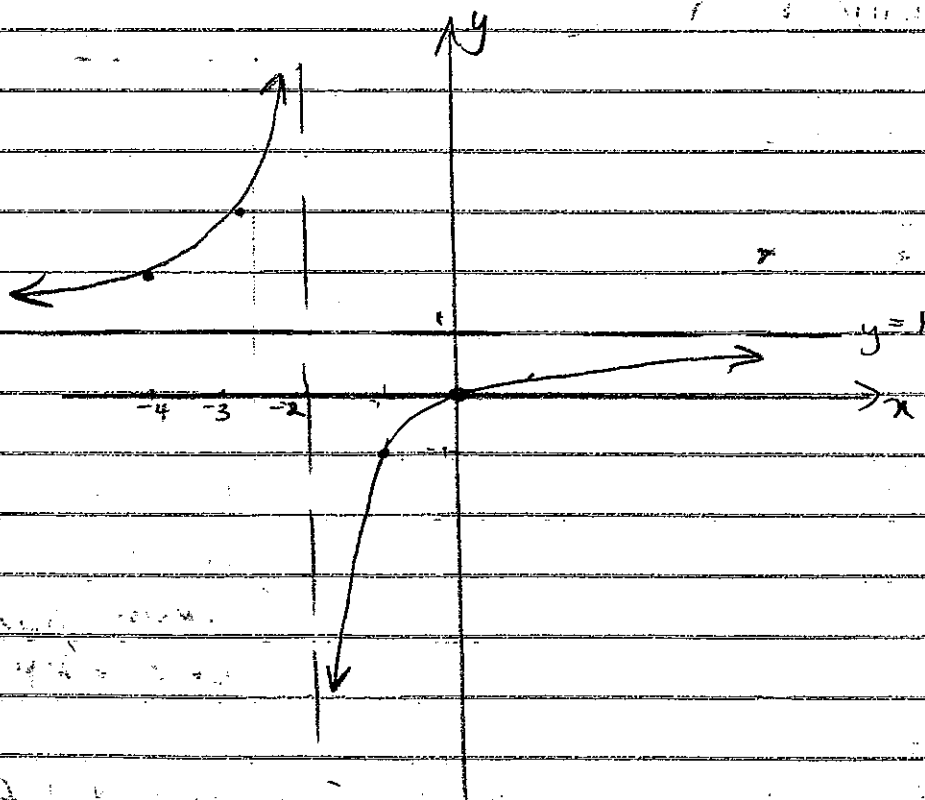
Award 1

(ii) horizontal asymptote  $y=1$

MUST HAVE  
EQUATION

(iii)  $x+2 \neq 0 \therefore x \neq -2$

Award 1



MUST USE  
RULER &  
HAVE CONSISTENT  
SCALE

$$a) \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = n$$

1. Show result true for  $n=1$

$$\text{ie LHS} = \frac{1}{3} \quad \text{RHS} = \frac{1}{2 \times 1 + 1}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$  result true for  $n=1$

2. Assume result true for  $n=k$

$$\text{ie } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} = k$$

3. Show true for  $n=k+1$

$$\text{ie } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = k+1$$

$$\text{ie } \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$$

$$\text{LHS} = \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$$

$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2k+3}$$

$$= \text{RHS}$$

result true for  $n=k+1$

As result is true for  $n=1$ , then result must be true for  $n=1+1$ , ie  $n=2$ , etc

Award 1. If student has written 'n=k' award 0

Award 1 for correct substitution assumption statement into show statement

Award 1 for correct resolution of LHS = RHS

Award 1 for correct statement

b)  $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$   
 show true for  $n = k+1$

If  $n = k+1$  then

$$1 + 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1)2^{k+1-1} = 1 + (k+1-1) \cdot 2^{k+1}$$

$$= 1 + k \cdot 2^{k+1}$$

For -

$$1 + (k-1)2^k + (k+1)2^k = 1 + k \cdot 2^{k+1}$$

Award 1.

LHS

$$= 1 + k \cdot 2^k - 2^k + k \cdot 2^k + 2^k$$

$$= 1 + 2k \cdot 2^k$$

$$= 1 + k \cdot 2^1 \cdot 2^k$$

$$= 1 + k \cdot 2^{k+1}$$

$$= \text{RHS}$$

Award 1

for this step

MUST have

EVERY STEP.

Watch for fudging